Data Structures & Algorithms Problem Set

20 Core Practice Problems • Arrays, Hashing, Two Pointers, Dynamic Programming & Intervals

1. Two Sum with Indices

ARRAYS & HASHING

Given an integer array and a target value, find two distinct elements whose sum is exactly equal to the target. Return their zero-based indices. The same array position cannot be used twice. Exactly one valid pair is guaranteed.

INPUT FORMAT

Line 1: `N T`
Line 2: `N` space-separated integers.

OUTPUT FORMAT

Print the two zero-based indices in increasing order.

SAMPLE INPUT

```
6 9
2 7 11 15 3 4
```

SAMPLE OUTPUT

```
0 1
```

Explanation: A[0] + A[1] = 2 + 7 = 9.

Constraints: $1 \le N \le 2 \times 10^5$, $-10^9 \le A[i], T \le 10^9$

2. Longest Consecutive Sequence

HASHING

Given an unsorted array of integers, find the length of the longest sequence of consecutive integer values. The elements do not need to be adjacent in the original array. Duplicate values do not increase the sequence length.

INPUT FORMAT

Line 1: `N`
Line 2: `N` integers.

OUTPUT FORMAT

Print the length of the longest consecutive sequence.

SAMPLE INPUT

```
6
100 4 200 1 3 2
```

SAMPLE OUTPUT

```
4
```

Explanation: 1, 2, 3, 4 form the longest consecutive sequence.

Constraints: $1 \le N \le 2 \times 10^5$, $-10^9 \le A[i] \le 10^9$

3. Subarray Sum Equals K

PREFIX SUM & HASHING

Given an integer array and an integer K, count the number of contiguous, non-empty subarrays whose sum is exactly K. The array may contain positive, negative, and zero values.

INPUT FORMAT

Line 1: `N K`
Line 2: `N` integers.

OUTPUT FORMAT

Print the total number of qualifying subarrays.

SAMPLE INPUT

```
5 3
1 2 1 1 1
```

SAMPLE OUTPUT

```
3
```

Explanation: [1,2], [2,1], and [1,1,1] each have sum 3.

Constraints: $1 \le N \le 2 \times 10^5$, $-10^9 \le A[i], K \le 10^9$

4. Longest Zero-Sum Subarray

PREFIX SUM

Find the maximum length of a contiguous, non-empty subarray whose elements sum to zero.

INPUT FORMAT

Line 1: `N`
Line 2: `N` integers.

OUTPUT FORMAT

Print the maximum length. Print 0 if no valid subarray exists.

SAMPLE INPUT

```
6
15 -2 2 -8 1 7
```

SAMPLE OUTPUT

```
5
```

Explanation: [-2, 2, -8, 1, 7] has sum 0 and length 5.

Constraints: $1 \le N \le 2 \times 10^5$, $-10^9 \le A[i] \le 10^9$

5. Count Subarrays with XOR K

PREFIX XOR

Given an array of non-negative integers and a target K, count the number of contiguous, non-empty subarrays whose bitwise XOR is exactly K.

INPUT FORMAT

Line 1: `N K`
Line 2: `N` integers.

OUTPUT FORMAT

Print the number of qualifying subarrays.

SAMPLE INPUT

```
5 6
4 2 2 6 4
```

SAMPLE OUTPUT

```
2
```

Explanation: [4,2] and [2,6] have XOR equal to 6.

Constraints: $1 \le N \le 2 \times 10^5$, $0 \le A[i], K < 2^{20}$

6. Maximum Product Subarray

DYNAMIC PROGRAMMING

Find the maximum product obtainable from any non-empty contiguous subarray. The array may contain positive values, negative values, and zero.

INPUT FORMAT

Line 1: `N`
Line 2: `N` integers.

OUTPUT FORMAT

Print the maximum product.

SAMPLE INPUT

```
5
2 3 -2 4 -1
```

SAMPLE OUTPUT

```
48
```

Explanation: The entire array has product 48.

Constraints: $1 \le N \le 2 \times 10^5$, $-10 \le A[i] \le 10$

7. Product of Array Except Self

PREFIX & SUFFIX PRODUCTS

For every position i, calculate the product of all array elements except A[i]. Division is not allowed.

INPUT FORMAT

Line 1: N
Line 2: N integers.

OUTPUT FORMAT

Print N products in the original order.

SAMPLE INPUT

```
4
1 2 3 4
```

SAMPLE OUTPUT

```
24 12 8 6
```

Explanation: For index 0, product of all other elements is 2 × 3 × 4 = 24.

Constraints: $2 \le N \le 2 \times 10^5$, $-20 \le A[i] \le 20$

8. Majority Element II

ARRAY / VOTING

Given an integer array, find every value occurring strictly more than floor(N/3) times. At most two values can satisfy this condition.

INPUT FORMAT

Line 1: N
Line 2: N integers.

OUTPUT FORMAT

Print qualifying values in increasing order. Print -1 if none qualifies.

SAMPLE INPUT

```
8
3 2 3 2 2 1 2 3
```

SAMPLE OUTPUT

```
2 3
```

Explanation: 2 occurs 4 times and 3 occurs 3 times; both > floor(8/3)=2.

Constraints: $1 \le N \le 2 \times 10^5$

9. Missing and Repeating Number

ARRAYS

An array should contain every integer from 1 through N exactly once. One value appears twice and another value is missing. Find both.

INPUT FORMAT

Line 1: N
Line 2: N integers from 1 to N.

OUTPUT FORMAT

Print the repeated value followed by the missing value.

SAMPLE INPUT

```
5
1 2 2 5 4
```

SAMPLE OUTPUT

```
2 3
```

Explanation: 2 is repeated and 3 is missing.

Constraints: $1 \le N \le 2 \times 10^5$

10. Maximum Subarray with One Deletion

DYNAMIC PROGRAMMING

Find the maximum sum of a non-empty contiguous subarray when at most one element may be deleted. The deleted element contributes nothing to the sum.

INPUT FORMAT

Line 1: `N`
Line 2: `N` integers.

OUTPUT FORMAT

Print the maximum achievable sum.

SAMPLE INPUT

```
4
1 -2 0 3
```

SAMPLE OUTPUT

```
4
```

Explanation: Deleting -2 leaves [1,0,3], whose sum is 4.

Constraints: $1 \le N \le 2 \times 10^5$, $-10^9 \le A[i] \le 10^9$

11. Count Distinct Values in Every Window

SLIDING WINDOW

Given an array and a window size K, count the number of distinct values in every contiguous window of exactly K elements.

INPUT FORMAT

Line 1: `N K`
Line 2: `N` integers.

OUTPUT FORMAT

Print N-K+1 distinct counts.

SAMPLE INPUT

```
7 3
1 2 1 3 4 2 3
```

SAMPLE OUTPUT

```
2 3 3 3 3
```

Explanation: The first window [1,2,1] contains 2 distinct values.

Constraints: $1 \le K \le N \le 2 \times 10^5$

12. Longest Subarray with At Most K Distinct Values

SLIDING WINDOW

Find the maximum length of a contiguous subarray containing at most K distinct values.

INPUT FORMAT

Line 1: `N K`
Line 2: `N` integers.

OUTPUT FORMAT

Print the maximum valid length. If K=0, print 0.

SAMPLE INPUT

```
7 2
1 2 1 2 3 2 2
```

SAMPLE OUTPUT

```
4
```

Explanation: [1,2,1,2] contains exactly 2 distinct values and has length 4.

Constraints: $1 \le N \le 2 \times 10^5$, $0 \le K \le N$

13. Top K Frequent Elements

HASHING & HEAP

Given an integer array, find the K values with the highest frequencies. If two values have the same frequency, the smaller value must come first.

INPUT FORMAT

Line 1: `N K`
Line 2: `N` integers.

OUTPUT FORMAT

Print exactly K values in decreasing frequency order.
Resolve ties by increasing value.

SAMPLE INPUT

```
7 2
1 1 1 2 2 3 4
```

SAMPLE OUTPUT

```
1 2
```

Explanation: 1 occurs 3 times and 2 occurs 2 times.

Constraints: $1 \le K \le$ number of distinct values $\le N \le 2 \times 10^5$

14. Longest Arithmetic Subarray

ARRAYS

Find the length of the longest contiguous subarray where the difference between every pair of consecutive elements is the same.

INPUT FORMAT

Line 1: `N`
Line 2: `N` integers.

OUTPUT FORMAT

Print the maximum length.

SAMPLE INPUT

```
7
10 7 4 1 3 5 7
```

SAMPLE OUTPUT

```
4
```

Explanation: [10,7,4,1] has common difference -3.

Constraints: $1 \le N \le 2 \times 10^5$

15. Maximum Circular Subarray Sum

KADANE'S ALGORITHM

Treat the array as circular, meaning the last element is adjacent to the first. Find the maximum sum of a non-empty contiguous subarray, including a subarray that wraps around.

INPUT FORMAT

Line 1: `N`
Line 2: `N` integers.

OUTPUT FORMAT

Print the maximum circular subarray sum.

SAMPLE INPUT

```
5
5 -3 5 -2 4
```

SAMPLE OUTPUT

```
11
```

Explanation: The wrapping subarray [4,5,-3,5] has sum 11.

Constraints: $1 \le N \le 2 \times 10^5$, $-10^9 \le A[i] \le 10^9$

16. Count Pairs with Difference K

HASHING

Count unordered pairs of distinct indices whose values differ by exactly K. Each pair of indices must be counted once.

INPUT FORMAT

Line 1: `N K`
Line 2: `N` integers.

OUTPUT FORMAT

Print the total number of valid index pairs.

SAMPLE INPUT

```
5 2
1 3 5 3 1
```

SAMPLE OUTPUT

```
4
```

Explanation: Four distinct index pairs have difference 2.

Constraints: $1 \le N \le 2 \times 10^5$, $0 \le K \le 10^9$

17. Smallest Missing Positive

ARRAYS / IN-PLACE

Find the smallest positive integer that does not appear in the array. The array may contain negative values, zero, duplicates, and values larger than N.

INPUT FORMAT

Line 1: `N`
Line 2: `N` integers.

OUTPUT FORMAT

Print the smallest missing positive integer.

SAMPLE INPUT

```
5
3 4 -1 1 2
```

SAMPLE OUTPUT

```
5
```

Explanation: 1, 2, 3, 4 are present, so 5 is the smallest missing positive.

Constraints: $1 \le N \le 2 \times 10^5$, $-10^9 \le A[i] \le 10^9$

18. Merge Overlapping Intervals

INTERVALS / SORTING

Given N closed intervals [start,end], merge every pair of intervals that overlap. The resulting intervals must be non-overlapping and sorted by starting value.

INPUT FORMAT

Line 1: `N`
Next N lines: `start end`

OUTPUT FORMAT

Print the merged intervals, one per line.

SAMPLE INPUT

```
4
1 3
2 6
8 10
9 12
```

SAMPLE OUTPUT

```
1 6
8 12
```

Explanation: [1,3] and [2,6] merge into [1,6]; [8,10] and [9,12] merge into [8,12].

Constraints: $1 \le N \le 2 \times 10^5$, start ≤ end

19. Insert Interval

INTERVALS

You are given sorted, non-overlapping intervals and one new interval. Insert the new interval while preserving the sorted and non-overlapping properties. Merge every interval that overlaps the new interval.

INPUT FORMAT

Line 1: `N`
Next N lines: existing intervals
Final line: `new_start new_end`

OUTPUT FORMAT

Print the resulting intervals, one per line.

SAMPLE INPUT

```
3
1 2
5 7
9 12
6 10
```

SAMPLE OUTPUT

```
1 2
5 12
```

Explanation: [6,10] overlaps [5,7] and [9,12], merging into [5,12].

Constraints: $0 \le N \le 2 \times 10^5$

20. Best Time to Buy and Sell Stock

GREEDY / ARRAYS

Given the price of a stock for each day, choose exactly one day to buy and a later day to sell. Maximize the profit from this single transaction. If no profitable transaction is possible, return 0.

INPUT FORMAT

Line 1: `N`
Line 2: `N` daily prices.

OUTPUT FORMAT

Print the maximum possible profit.

SAMPLE INPUT

```
6
7 1 5 3 6 4
```

SAMPLE OUTPUT

```
5
```

Explanation: Buy at 1 and sell at 6 (profit = 5).

Constraints: $2 \le N \le 2 \times 10^5$, $0 \le price[i] \le 10^9$